Beam search

QUEUE ←path only containing root
WIDTH ←<number>

WHILE QUEUE is not empty
AND goal is not reached
DO remove all paths from the QUEUE
create new paths (to all children)
reject the new paths with loops
sort the new paths (by heuristic)
add the WIDTH best paths to QUEUE

IF goal reached
THEN success
ELSE failure

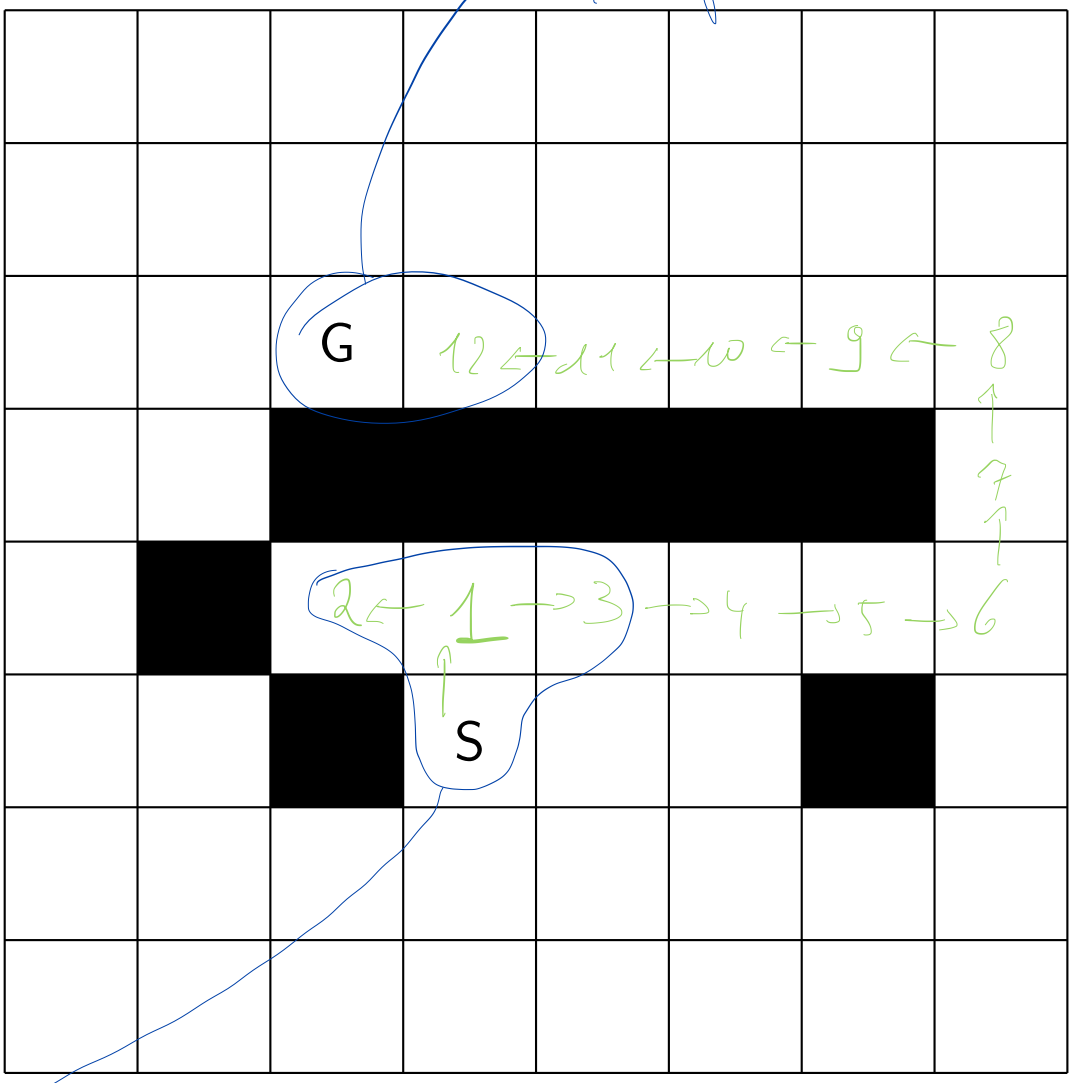
Optimisation : before the last DO-line :
"remove the paths with no successors"

Path Search: Manhattan Distance

4	3	2	3	4	5	6	7
3	2	1	2	3	4	5	6
2	1	G 0	1	2	3	4	5
3	2						6
4		2	3	4	5	6	7
5	4		S 4	5	6		8
6	5	4	5	6	7	8	9
7	6	5	6	7	8	9	10

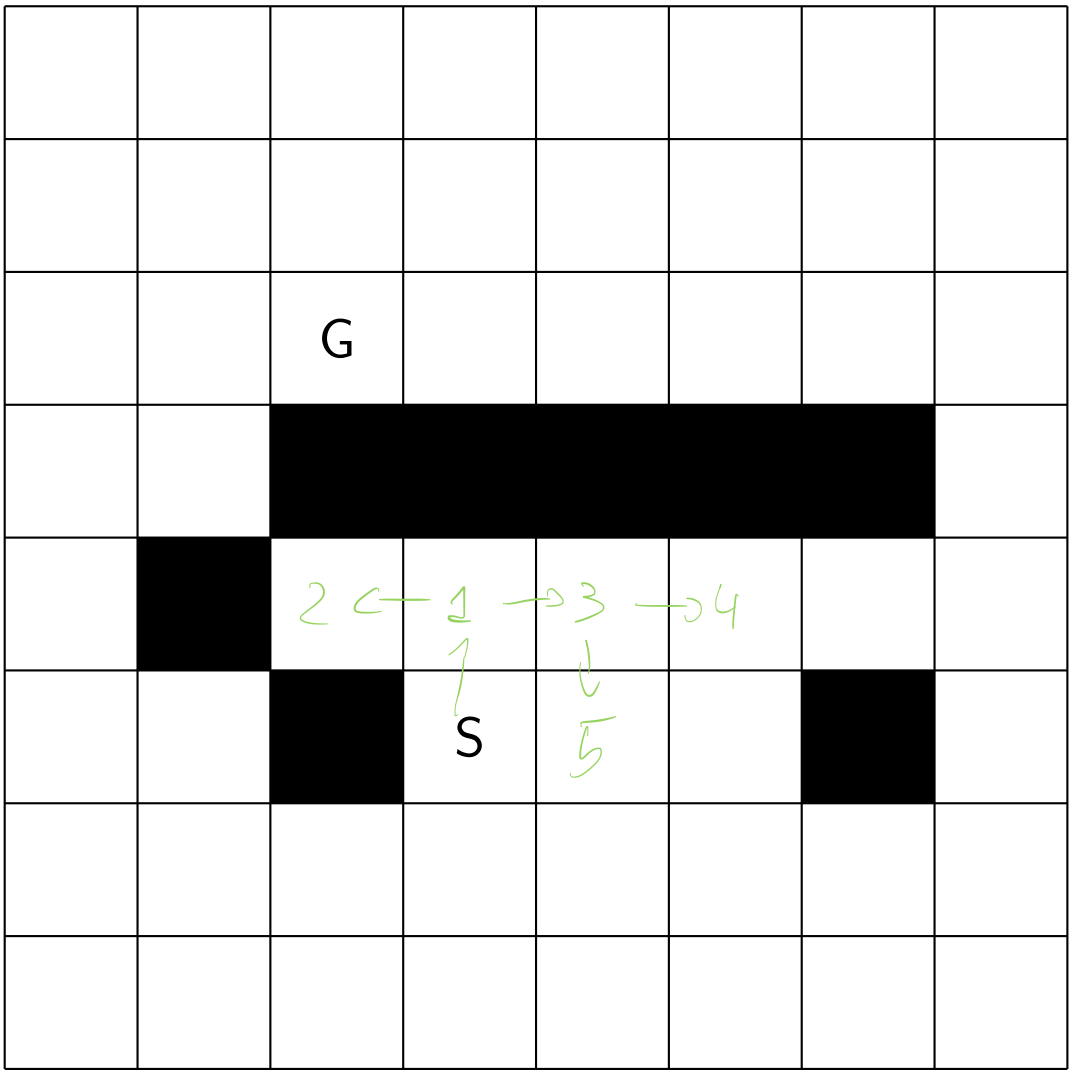
Path Search: hill climbing

geen pijl van 12 naar G, want na genereren en controleren van kinderen van 12 zit een pad naar G in QUEUE, dus STOP



- na genereren kinderen pad (S 1), 2 kinderen in QUEUE, nl. (S 1 2) en (S 1 3).
- enig kind van pad (S 1 2) is : (S 1 2 1), maar dat genereert loop, dus reject van (S 1 2 1)
- verder gaan met volgend pad in QUEUE, nl. (S 1 3), ...

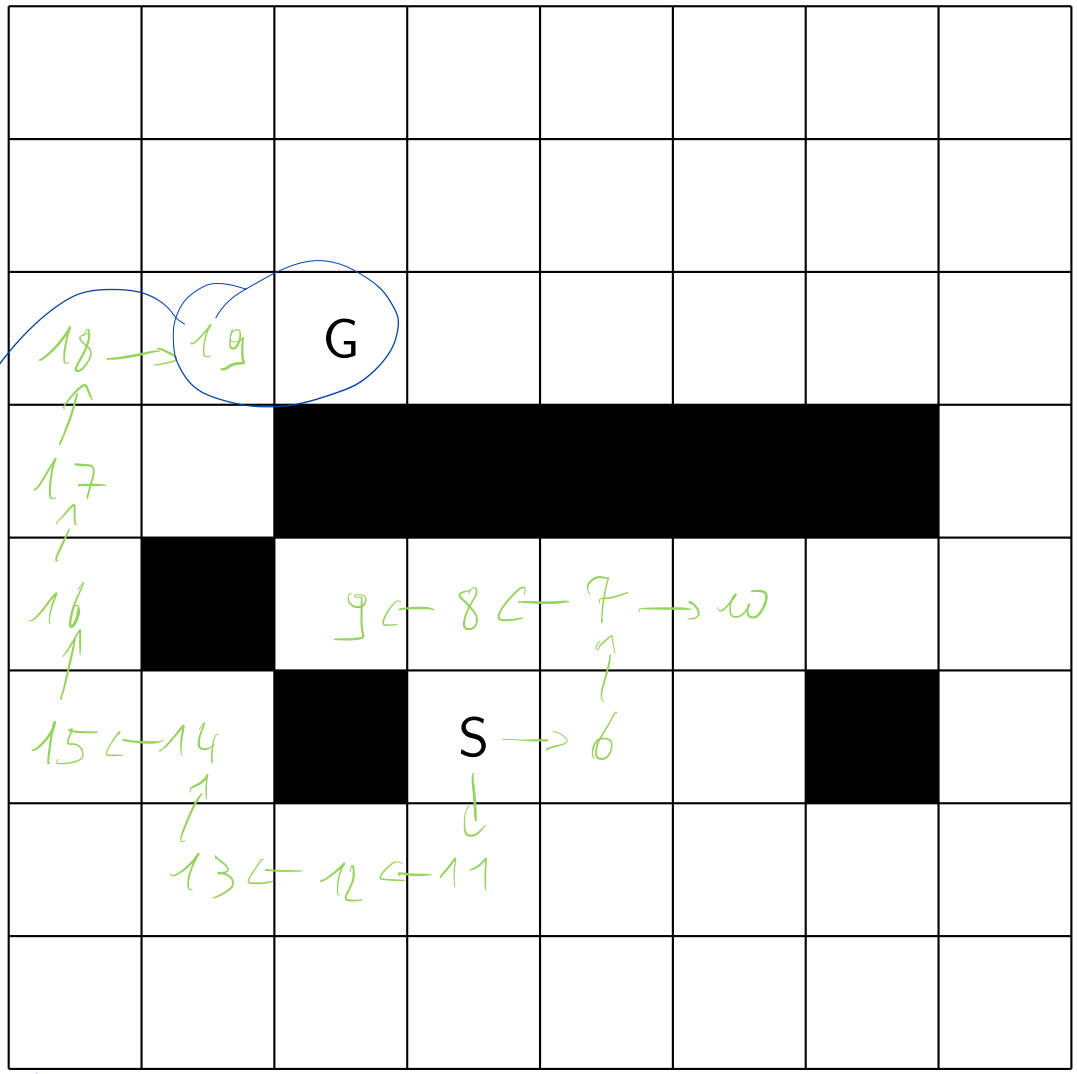
Path Search: greedy search (1)



Voor de duidelijkheid start ik opnieuw vanaf de zesde stap op een nieuwe figuur.

Path Search: greedy search (2)

Merk op dat een aantal plaatsen indefinitief meermaals bezocht worden!



hier weer geen pijl: zie hill-climbing voor reden

Water Jugs

Representation: $[x, y]$ with

- x contents of 4l jug
- y contents of 3l jug

Begin: $[0, 0]$
End: $[2, 0]$

Rules:

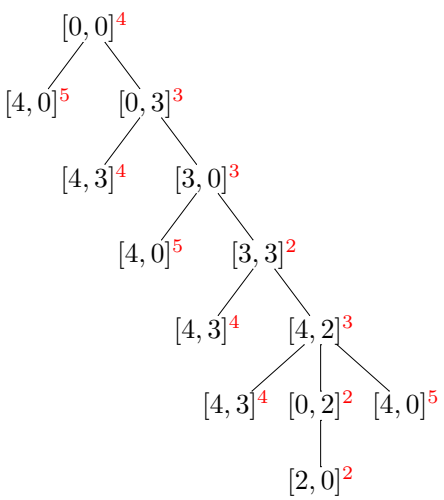
$$\begin{cases} [x,y], x < 4 & \rightarrow & [4, y] \\ [x,y], y < 3 & \rightarrow & [x, 3] \\ [x,y], x > 0 & \rightarrow & [0, y] \\ [x,y], y > 0 & \rightarrow & [x, 0] \\ [x,y], x + y > 4 \,\&\, y > 0 & \rightarrow & [4, y - (4 - x)] \\ [x,y], x + y \leq 4 \,\&\, y > 0 & \rightarrow & [x + y, 0] \\ [x,y], x + y > 3 \,\&\, x > 0 & \rightarrow & [x - (3 - y), 3] \\ [x,y], x + y \leq 3 \,\&\, x > 0 & \rightarrow & [0, x + y] \end{cases}$$

Water Jugs: Heuristic

$$h([x, y]) = f(x) + f(y)$$

with

x	0	1	2	3	4
$f(x)$	2	1	0	1	3

"To get a 2, you need a 1 or a 3 and 0 is better than 4."

Water Jugs: Hill Climbing